

NLPizza Summary Report:

Silvia Bonomi, Samuele D’Avenia, Andrea Nugara, Maria Pronestì

1 Introduction to task and data

The aim of this project is to analyse the task of Relation Extraction (RE), meaning that given a sentence, the entities should be identified, and then a relationship is assigned between them. The focus is on the second part of the task, i.e. correctly classifying the relations using both classical machine learning (ML) and Deep Learning (DL) methods.

Dataset For the project two datasets are used: TACRED by [Zhang et al. \[2017\]](#), along with its revisited version called TACREV by [Alt et al. \[2020\]](#). The former is constructed using crowd-sourcing, while for the latter the most challenging instances from the dev and test set were relabelled by linguistic experts. For this reason the validation and test set from TACREV are used, as they are an improved version of the former.

Both have the same structure and the features that are used in the models are briefly described here. Words that are the subject and object of the sentence have an attribute specifying their type. For all words we also have the associated Part-of-Speech (POS) and Named-Entity-Recognition (NER) tags, obtained by using the StanfordCoreNLP tool by [Manning et al. \[2014\]](#).

Pre-processing Classical pre-processing steps for NLP tasks include stop-words removal and lemmatization and/or stemming. However, for the task of relation extraction these may cause the removal of key elements for understanding relation types, and for this reason are not performed.

2 Approaches considered

Regular Expression An initial attempt considers the development of hand-crafted regex patterns for some of the relations. While being extremely fast to execute, this approach is particularly time-consuming for the development of appropriate patterns and the results obtained are not satisfactory.

Random Forest (RF) For the RF model the features used are described. First a sentence embedding is obtained by using the average of the individual word embeddings from a pre-trained FastText model by [Bojanowski et al. \[2016\]](#). Moreover, the distance between subject and object entities, their similarity, the object and subject type and POS are included. Hyper-parameter tuning is performed on the number of estimators and the maximum depth of the tree. Also note that due to the unbalanced nature of the dataset, class re-weighting is applied.

Position-Aware LSTM The first DL approach was a very simple LSTM model, which had unsatisfactory results. A more clever model, which is considered here, is the one by [Zhang et al. \[2017\]](#). In a very simplified way it consists of an LSTM where the input are words enriched with POS and NER embedding. Additionally, a positional attention mechanism is employed to obtain a sentence level representation from the word embeddings obtained with the LSTM layer.

3 Results

All the metrics reported are obtained using micro-averaging, as it is present in the literature for this dataset. Note that, this metric does not reward the model for correctly predicting *no_relation*. Precision, Recall and F1 obtained this way are shown in Table 1. The right side of the table contains the same metrics but only considering the 5 most frequent relations in the training set.

Model	All Relations			Most frequent relations		
	Precision	Recall	F1	Precision	Recall	F1
PA-LSTM	69.0%	69.5%	69.2%	78.7%	78.3%	78.5%
RF	46%	49%	48%	47%	75%	57%

Table 1: Performance Metrics for RF and PA-LSTM Models

As expected the DL model outperforms the RF, however note that while the training of the RF took approximately 5 minutes, for the PA-LSTM model this was 8 hours (both on *Intel Core i5* processor). For both models some relations were never predicted correctly, however the RF had a larger number of those.

Error Analysis The first part of the error analysis tried to identify relation-agnostic aspects of the sentences that affect the performance. The sentence length along with the distance between the entities is considered. For both models the performance degraded greatly when the distance between the entities increased, while the sentence length had a smaller effect.

For the second part of the error analysis the goal is to explore whether the model could correctly distinguish between relations of similar meaning. From this exploration, it is observed that the most common error for both models is the prediction of *no_relation* for other relations and vice-versa. Almost all other errors are caused by the model outputting a relation falling in the same group of similar ones. However, note that these are also the most common errors identified by the linguistic experts as pointed out in Alt et al. [2020]. Since the training set was not re-labelled these errors are likely to be present. This potentially points to the model learning on wrong sentences, which may degrade its performance. During the error analysis, it also appears that there is a class of sentences which are wrongly classified due to mislabelled subject or object type. This is problematic as the model requires this information to classify some relations correctly.

4 Conclusion

The models considered were analyzed with a focus on a quantitative and qualitative error analysis, which helped in identifying some issues. The biggest issue appears to be that both models perform poorly with long distance between the entities and for this type of issues transformers with their self-attention mechanism could be beneficial. Indeed, the SOTA architectures for this task are from this family. As further expansions of the error analysis it would be interesting to try and quantify the errors which are due to the wrong NER labelling. While performing the error analysis, it was noted that some relations barely occur in the test set. For this reason, enriching the test set with more examples containing relations of those types would be useful to obtain a reliable estimate of the effectiveness of the model on those.

References

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